Integral Transforms & Their Applications

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Abstract

Integral transforms convert differential and integral equations into algebraic equations, simplifying the solution process. This project provides a study of four different integral transforms – the Laplace, Fourier, Mellin, and Weierstrass transforms – examining definitions, key properties, and applications to ordinary and partial differential equations.

The Laplace transform is presented as a method for solving initial value problems, with its convolution theorem applied to Volterra-type integral equations and transfer functions in linear dynamical systems. The Fourier transform is developed from Fourier series and applied to boundary value problems and partial differential equations, including the heat equation and Laplace's equation. The Mellin transform is introduced in the context of scaling and multiplicative structure, with its connections to both the Laplace and Fourier transforms via a logarithmic change of variables. Finally, the Weierstrass transform is studied as a Gaussian smoothing operator, interpreted as the heat semigroup, and characterised through its frequency-domain behaviour.

A comparative investigation of all four transforms is provided, highlighting how the choice of kernel and domain determines the class of problems each transform is best suited to address. The central conclusion is that transform selection should be guided by the dominant structure of the problem, whether it is characterised by time-shifts, periodicity, scaling, or diffusion.

1 Introduction

1.1 Background and Motivation

1.1.1 Motivation

Integral transforms are mathematical techniques that convert differential (and often times integral) equations into algebraic equations. The concept was first introduced by Leonhard Euler in 1763 and has since been developed extensively. A wide variety of transforms have been proposed, each designed to address specific mathematical structures and application areas. Today, they form a central part of applied mathematics, continuing to play an important role in the analysis of both differential and integral equations.

Integral transforms are particularly useful because algebraic equations are generally easier to manipulate than their differential counterparts. Initial and boundary conditions can often be incorporated directly into the transformed equations, further reducing the complexity of finding solutions.

A further advantage is the treatment of convolution integrals, which are often analytically challenging when handled directly. By the nature of integral transforms, reducing operations into simple algebraic multiplication, convolution integrals also become simple products. Once this algebraic equation has been solved, the inverse transform is applied to recover the solution in the original domain.

Beyond differential equations, integral transforms also play an important role in signal processing, probability theory, asymptotic analysis, and the study of scaling phenomena in applied mathematics.

For further reading on integral transforms, see Debnath and Bhatta[I] and Sneddon[II] .

1.1.2 Limitations of Direct Methods for Solving Differential and Integral Equations

Direct methods for solving differential and integral equations have limitations, particularly when dealing with linear systems involving memory effects[III], delay terms, or convolution integrals. Direct analytical techniques are generally restricted to simpler problems and become impractical as the complexity of the equation increases. In applied science and engineering, integral equations frequently appear in *Convolution Form*[IV], which can become algebraically complicated and make direct solution approaches computationally demanding.

Integral transforms, therefore, provide an alternative by preserving the linearity of the system while converting convolution integrals into simple algebraic products.

[I] Lokenath Debnath and Dambaru Bhatta. *Integral Transforms and Their Applications*. CRC Press, 2007.

[II] I.N. Sneddon. "The Use of Integral Transforms". In: *MvGraw Hill Book Company* (1972).

[III] A *Memory Effect* is the situation where the current state or evolution of a dynamical system depends not only on the present values but on the entire past history of the system (Malgorzata Peszyńska, *Analysis of an Integro-Differential Equation Arising from Modelling of Flows with Fading Memory Through Fissured Media*, 1995, https://sites.science.oregonstate.edu/~mpesz/documents/publications/P95a.pdf)

[IV] A *Convolution Integral* typically has the form

$$(f * g)(t) = \int_0^t f(t-\tau)g(\tau)\,d\tau.$$

1.1.3 General Idea of Solving in a Transformed Domain and Inverting the Solution

The idea behind using integral transforms to solve differential and integral equations is conceptually straightforward. Rather than solving a differential equation directly for an unknown function f, which can be difficult, the equation is first transformed into a different domain using an integral transform. This process converts the original differential or integral equation into an algebraic equation for the transformed function, typically denoted by F.

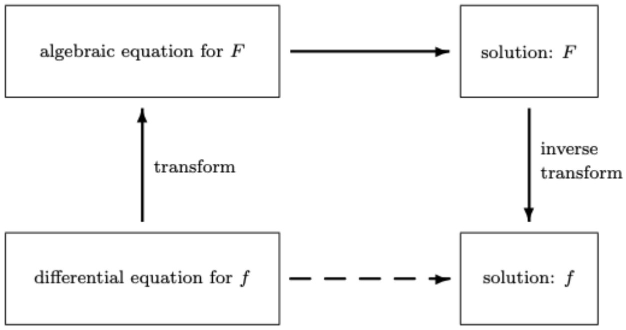

Figure 1.1: Steps showing the process of solving a differential equation f by means of a transform, particularly an integral transform

This procedure is shown in Figure 1 (Fraser[V]). Ideally, we would like to follow the direct path from the differential equation to its solution in the original domain. However, this direct approach is often analytically challenging. Instead, the transform-based approach provides an alternative route where we transform the original equation, solving the simpler algebraic problem, and then invert the solution. This indirect route is typically more systematic and efficient than attempting to solve the original problem directly.

1.2 Integral Transforms

1.2.1 Structure and Explanation of Integral Transforms

An integral transform is an operator that maps a function from its original domain into a new function in a transformed domain using an integral. This is done by combining the original function with a *kernel* (Weisstein[VI]), which determines how the transformation works. In general, an integral transform can be written in the form

$$F(s) = \int_a^b k(t, s) f(t)\, dt \tag{1.1}$$

where the function F is the transformed function defined on the frequency domain with the independent variable s, while f is the original function. The kernel k defines how the transformation is applied, while the limits of integration a and b (which could be finite or infinite) depend on the specific transform being used.

During the transformation, the original function $f(t)$ is multiplied by the kernel k and integrated over t. The kernel therefore acts as a weighting function, determining which parts of $f(t)$ contribute more or less strongly to the transformed function $F(s)$. In this way, the kernel shapes

[V] J. M. Fraser. *Integral Transforms Notes.* URL: https://webhomes.maths.ed.ac.uk/~jmf/Teaching/MT3/IntegralTransforms.pdf.

[VI] Eric W. Weisstein. *Integral Equation.* URL: https://mathworld.wolfram.com/IntegralEquation.html.

how information from the original domain is transferred and emphasised in the new domain.

The choice of kernel reflects the mathematical or physical structure of the system. For example, some kernels emphasise oscillatory behaviour, allowing periodic or wave-like components of the function to be identified, while others emphasise growth or decay, making them suitable for problems involving exponential behaviour.

Integral transforms are linear operators, preserving linear combinations of functions and enabling superposition principles to be applied before and after transformation.

For an integral transform to be useful, an inverse transform must exist. The inverse transform allows the transformed function to be mapped back to the original domain once a solution has been obtained. Thus, the transform and its inverse form a pair that enables problems to be reformulated, analysed, and ultimately expressed in terms of the original function (Davies[VII]).

For such an inverse to exist, the transform must be bijective – i.e. injective, to provide uniqueness; and surjective, such that every function in the transformed domain corresponds to some original function – to ensure that the inverse is well-defined.

1.2.2 Different Types of Integral Equations

An *integral equation* is an equation in which the unknown function to be determined appears under the integral sign.

Integral equations are classified according to the integration limits and position of the unknown function. If the limits of integration are fixed, the equation is called a *Fredholm integral equation*. If one limit (usually the upper limit) is variable, it is called a *Volterra integral equation* (Weisstein[VIII]).

Depending on the position of the unknown function, equations are further classified as being of the *first kind*, where the unknown function appears only under the integral sign, or of the *second kind*, where the unknown function appears both inside and outside the integral.

The four main types can be written as follows, where f is the unknown function, g is a known forcing function, and K is the kernel:

- **Fredholm integral equation of the first kind:**

$$g(x) = \int_a^b K(x,t)\, f(t)\, dt \tag{1.2}$$

- **Fredholm integral equation of the second kind:**

$$f(x) = g(x) + \int_a^b K(x,t)\, f(t)\, dt \tag{1.3}$$

- **Volterra integral equation of the first kind:**

$$g(x) = \int_a^x K(x,t)\, f(t)\, dt \tag{1.4}$$

[VII] B. Davies. *Integral Transforms and Their Applications*. Springer, 2002.
[VIII] Weisstein, see n. VI.

- **Volterra integral equation of the second kind:**

$$f(x) = g(x) + \int_a^x K(x,t)\, f(t)\, dt \tag{1.5}$$

The key difference between Fredholm and Volterra equations lies in the integration limits: Fredholm equations integrate over a fixed interval $[a, b]$, while Volterra equations have a variable upper limit x. This makes Volterra equations particularly suited to problems where the behaviour at a given point depends only on its past values, such as in initial value problems.

Equations of the first kind are generally more difficult to solve and more sensitive to data perturbations than those of the second kind, which have important implications for existence, uniqueness, and numerical stability.

1.2.3 Linearity of Integral Transforms

A fundamental property of integral transforms is *linearity*. This means that the transform preserves linear combinations of functions. More precisely, let $\mathcal{T}$ denote an integral transform defined by

$$\mathcal{T}[f](s) = \int_a^b k(t,s) f(t)\, dt.$$

Then for functions f, g and scalars α, β,

$$\mathcal{T}[\alpha f + \beta g](s) = \alpha \mathcal{T}[f](s) + \beta \mathcal{T}[g](s).$$

Proof.

$$\begin{aligned}
\mathcal{T}\left[\alpha f + \beta g\right](s) &= \int_a^b k(t,s)[\alpha f(t) + \beta g(t)]dt \\
&= \alpha \int_a^b k(t,s) f(t) dt + \beta \int_a^b k(t,s) g(t) dt \\
&= \alpha \mathcal{T}[f](s) + \beta \mathcal{T}[g](s).
\end{aligned}$$

□

Integral transforms of this standard form are linear due to the linearity of integration.

2 The Laplace Transform

The Laplace Transform was first used by the French Mathematician Pierre-Simon Laplace (1749–1827) and developed by the British physicist Oliver Heaviside (1850–1925), to simplify the solution of differential equations that describe physical processes (Britannica[I]). Today it is used most commonly by electrical engineers in the solution of various electronic circuit problems.

The Laplace transform maps a function of a real variable t into a function of a complex variable $s = \sigma + i\tau$. Under this transformation, differentiation becomes multiplication by s, converting differential equations into algebraic ones. The imaginary part τ represents oscillatory behaviour, while the real part σ governs exponential growth or decay.

[I] Encyclopaedia Britannica. *Laplace Transform.* URL: https://www.britannica.com/science/Laplace-transform.

2.1 The Laplace Transform as a Method for Solving ODEs

The Laplace transform was developed as a method for solving linear Ordinary Differential Equations (ODEs) with constant coefficients, specifically Initial Value Problems (IVPs). The Laplace transform converts *differentiation in the time domain* into *multiplication in the frequency domain* with initial conditions appearing as constants.

This transform-solve-invert procedure is suited to IVPs because the initial conditions at $t = 0$ appear directly as constants in the algebraic equation. In contrast, Boundary Value Problems (BVPs), where conditions are specified at two different points, are generally less amenable to the Laplace transform, as the transform is inherently defined on the half-line $t \geq 0$ and does not accommodate conditions imposed at later times.

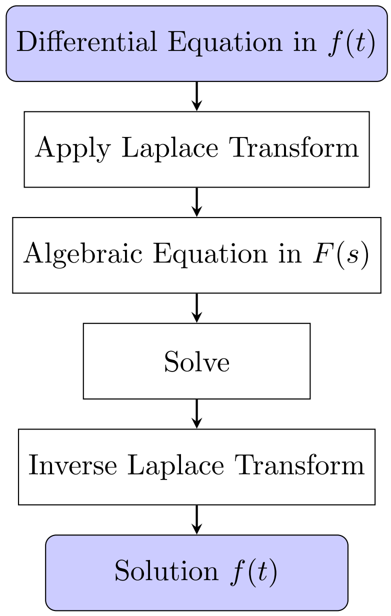

Figure 2.1: The transform-solve-invert procedure for initial-value problems using the Laplace transform

2.2 Definitions

Definition 2.2.1 (Locally Integrable Functions)**.** Let $I \subset \mathbb{R}$ be an open finite interval. A function $f : I \to \mathbb{C}$ is said to be *locally integrable* on I if

$$\int_I f(t)\,dt \quad \text{exists} \quad \text{and} \quad \int_I |f(t)|\,dt < \infty.$$

This condition ensures that the integral defining the Laplace transform is well-defined and finite.

Definition 2.2.2 (One-sided Laplace Transform)**.** Let $f : \mathbb{R}_+ \to \mathbb{C}$ be a locally integrable function, where $\mathbb{R}_+ = [0, \infty)$. The *one-sided Laplace transform* of f is defined by

$$\mathcal{L}\{f(t)\}(s) = F(s) := \int_0^\infty f(t)e^{-st}\,dt, \quad s \in \mathbb{C}.$$

In applications, t usually represents time and s is referred to as the *complex frequency*.

Definition 2.2.3 (Exponential Order)**.** A function $f : \mathbb{R}_+ \to \mathbb{C}$ is said to be of *exponential order* if there exist constants $M > 0$ and $\alpha \in \mathbb{R}$ such that

$$|f(t)| \leq Me^{\alpha t} \quad \text{for all } t \geq 0.$$

This condition restricts the growth of f.

Definition 2.2.4 (Region of Convergence (RoC))**.** The RoC of the Laplace transform $F(s)$ of a function $f(t)$ is the set of all complex numbers $s \in \mathbb{C}$ for which the integral

$$\int_0^\infty f(t)e^{-st}\,dt \quad \text{converges.}$$

For functions that are zero for $t < 0$, the RoC is always a right half-plane, which is crucial because the same algebraic expression for $F(s)$ can represent different time-domain functions (stable decaying versus unstable growing) depending on the chosen RoC (Baraniuk et al.[II]).

The Laplace transform does not always converge, even for functions that appear well-behaved. For example, functions such as $f(t) = e^{t^2}$ or $f(t) = \Gamma(t)$ grow faster than the exponential and cause the integral to diverge for every $s \in \mathbb{C}$.

To guarantee convergence, sufficient conditions on f are needed, which are made precise in the following theorem.

Theorem 2.2.1 (Existence of the Laplace Transform)**.** *Let $f : \mathbb{R}_+ \to \mathbb{C}$ be locally integrable and of exponential order α, i.e. there exist constants $M, c > 0$ such that $|f(t)| \leq Me^{\alpha t}$ for all $t \geq c$. Then, for any $s \in \mathbb{C}$ satisfying $\Re(s) > \alpha$, the Laplace transform*

$$F(s) = \int_0^\infty f(t)e^{-st}\,dt$$

converges absolutely[III].

Proof. Suppose $|f(t)| \leq Me^{\alpha t}$ for all $t > c > 0$. Then

$$|f(t)e^{-st}| = |f(t)|e^{-\Re(s)t} \leq Me^{-(\Re(s)-\alpha)t}.$$

If $\Re(s) > \alpha$, the function $e^{-(\Re(s)-\alpha)t}$ decays exponentially as $t \to \infty$, and

$$\int_0^\infty |f(t)e^{-st}|\,dt \leq \int_0^\infty Me^{-(\Re(s)-\alpha)t}\,dt = \frac{M}{\Re(s)-\alpha} < \infty.$$

Hence, the Laplace integral converges absolutely whenever $\Re(s) > \alpha$. □

2.3 Key Properties

Theorem 2.3.1 (Laplace Transform of the First Derivative)**.** *Let $f : \mathbb{R}_+ \to \mathbb{C}$ be differentiable and suppose that both f and f' are locally integrable and of exponential orders $\alpha, \alpha' \in \mathbb{R}$ (in other words, there exits $M, M' > 0$ and $\alpha, \alpha' \in \mathbb{R}$ such that $|f(x)| \leq Me^{\alpha t}$ and $|f'(x)| \leq M'e^{\alpha' t}$). Denote*

$$F(s) = \mathcal{L}\{f(t)\}(s) = \int_0^\infty f(t)e^{-st}\,dt.$$

Then, for all $s \in \mathbb{C}$ with $\Re(s) > \max\{\alpha, \alpha'\}$,

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0).$$

Proof. Starting from the definition,

$$\mathcal{L}\{f'(t)\}(s) = \int_0^\infty f'(t)e^{-st}\,dt.$$

Integrate by parts with $u = e^{-st}$ and $dv = f'(t)\,dt$, so that $du = -se^{-st}\,dt$ and $v = f(t)$. This gives

$$\int_0^\infty f'(t)e^{-st}\,dt = \Big[f(t)e^{-st}\Big]_0^\infty + s\int_0^\infty f(t)e^{-st}\,dt.$$

[II]Richard Baraniuk et al. *Region of Convergence for the Laplace Transform*. Online: `https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Signal_Processing_and_Modeling/Signals_and_Systems_(Baraniuk_et_al.)/11%3A_Laplace_Transform_and_Continuous_Time_System_Design/11.06%3A_Region_of_Convergence_for_the_Laplace_Transform`. Accessed: 2026-04-04.

[III]An integral $\int_0^\infty f(t)\,dt$ converges absolutely if $\int_0^\infty |f(t)|\,dt < \infty$.

The boundary term at infinity vanishes because f is of exponential order α, meaning that for $\Re(s) > \max\{\alpha, \alpha'\} \geq \alpha$,

$$\lim_{t\to\infty} f(t)e^{-st} = 0.$$

Evaluating the remaining boundary term at $t = 0$ yields $-f(0)$. Therefore,

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0).$$

□

The above result can be extended to higher derivatives by induction.

Theorem 2.3.2 (Higher-Order Derivatives)**.** *Let $f : \mathbb{R}_+ \to \mathbb{C}$ be N-times differentiable and all derivatives up to order N are locally integrable and of exponential order (meaning that there exist constants $M_0, M_1, \ldots, M_N > 0$ and $\alpha_0, \alpha_1, \ldots, \alpha_N \in \mathbb{R}$ such that $\left|\frac{d^n f}{dt^n}\right| \leq M_n e^{\alpha_n t}$ for all $t \geq 0$ and $n = 0, 1, \ldots, N$). Then*

$$\mathcal{L}\{f^{(N)}(t)\}(s) = s^N F(s) - \sum_{n=0}^{N-1} s^{N-1-n} f^{(n)}(0).$$

Proof. We prove the result by induction on N.

Base Step:

From the first derivative property of the Laplace transform,

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0),$$

which agrees with the formula for $N = 1$:

$$s^1 F(s) - \sum_{n=0}^{0} s^0 f^{(n)}(0) = sF(s) - f(0).$$

Inductive Step:

Assume that the formula holds for some $N = k \geq 1$, i.e.

$$\mathcal{L}\{f^{(k)}(t)\}(s) = s^k F(s) - \sum_{n=0}^{k-1} s^{k-1-n} f^{(n)}(0).$$

We must show that it holds for $k + 1$.

Using the first derivative property applied to $f^{(k)}(t)$,

$$\mathcal{L}\{f^{(k+1)}(t)\}(s) = s\,\mathcal{L}\{f^{(k)}(t)\}(s) - f^{(k)}(0).$$

which holds since the function $f^{(k)}$ is of exponential order. By the inductive hypothesis,

$$\mathcal{L}\{f^{(k+1)}(t)\}(s) = s\left(s^k F(s) - \sum_{n=0}^{k-1} s^{k-1-n} f^{(n)}(0)\right) - f^{(k)}(0).$$

Expanding and combining the final term into the sum gives

$$\mathcal{L}\{f^{(k+1)}(t)\}(s) = s^{k+1}F(s) - \sum_{n=0}^{k} s^{k-n} f^{(n)}(0).$$

This is exactly the required formula for $k+1$:

$$\mathcal{L}\{f^{(k+1)}(t)\}(s) = s^{k+1}F(s) - \sum_{n=0}^{k} s^{k-n} f^{(n)}(0).$$

Hence, by induction, the result holds for all $N \in \mathbb{N}$. □

Theorem 2.3.3 (Initial Value Property)**.** *Let $f : \mathbb{R}_+ \to \mathbb{C}$ be locally integrable and of exponential order $\alpha \in \mathbb{R}$. Suppose further that f is continuous at $t = 0$ from the right, i.e. the limit*

$$f(0) := \lim_{t \to 0^+} f(t)$$

exists and is finite, and that f' is locally integrable and of exponential order. Let

$$F(s) = \mathcal{L}\{f(t)\}(s) = \int_0^\infty f(t)e^{-st}\,dt.$$

Then, for all $s \in \mathbb{C}$ with $\Re(s) > \alpha$,

$$\lim_{s \to +\infty} sF(s) = f(0).$$

Proof. From the first derivative formula,

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0). \tag{2.1}$$

Since f' is of exponential order, the integrand $f'(t)e^{-st}$ is bounded by a rapidly decaying exponential for all $t > 0$. Therefore, the left hand side is

$$\mathcal{L}\{f'(t)\}(s) = \int_0^\infty f'(t)e^{-st}\,dt \to 0 \quad \text{as } s \to \infty.$$

Rearranging the first derivative formula in Equation (2.1) and taking the limit as s tends to 0 gives

$$\lim_{s\to\infty} sF(s) = f(0) + \lim_{s\to\infty} \mathcal{L}\{f'(t)\}(s) = f(0).$$

□

The initial value property provides useful information about the behaviour of $F(s)$ as $|s| \to \infty$. However, to fully understand the relationship between a function and its Laplace transform, we need to establish a much stronger result, that the Laplace transform is *injective*. In other words, different functions cannot have the same Laplace transform. This uniqueness property is crucial for the inversion problem and justifies the use of the inverse Laplace transform.

The following theorems of injectivity and inversion have been adapted from Earl[IV] .

Theorem 2.3.4 (Injectivity of the Laplace Transform)**.** *Let $f : \mathbb{R}_+ \to \mathbb{R}$ be a continuous function*

[IV] Richard Earl. "Integral Transforms". Oxford Mathematical Institute. Lecture Notes. 2020.

of exponential order $\alpha \in \mathbb{R}$ such that $F(s) = \mathcal{L}\{f(t)\}(s) = 0$ for all $\Re(s) > \alpha$. Then $f = 0$ (Chierchia[V]).

Proof. We use the *Weierstrass Approximation Theorem*, which states that any continuous function on a closed, bounded interval can be uniformly approximated by a sequence of polynomials.

Let $k \in \mathbb{R}$ be a fixed value such that $k > \alpha$ and set $s = k + n + 1$ for some $n \geq 0$. Then

$$0 = \int_0^\infty f(t)e^{-st}\,dt = \int_0^\infty e^{-nt}\,e^{-kt}\,e^{-t}f(t)\,dt.$$

Make the change of variables $y = e^{-t}$ (so $t = -\log y$ and $dt = -dy/y$). The integral becomes

$$0 = \int_0^1 y^n\,y^k f(-\log y)\,dy = \int_0^1 y^n g(y)\,dy,$$

where $g(y) = y^k f(-\log y)$. Note that g is continuous on $(0, 1]$ and extends continuously to $y = 0$, since

$$|g(y)| = |y^k f(-\log y)| = e^{-kx}|f(x)| \leq Me^{(\alpha-k)x} \to 0 \quad \text{as } y \to 0^+ \ (x \to \infty).$$

By linearity, $\int_0^1 p(y)g(y)\,dy = 0$ for any polynomial $p : [0, 1] \to \mathbb{R}$. Let $\{p_n\}$ be a sequence of polynomials defined on $[0, 1]$ that converges uniformly to g (guaranteed by the Weierstrass Approximation Theorem). Then

$$\int_0^1 g(y)^2\,dy = \lim_{n\to\infty} \int_0^1 p_n(y)g(y)\,dy = 0.$$

Hence $g(y) = 0$ for all $y \in [0, 1]$. It follows that $f(x) = 0$ for all $x \geq 0$. □

Injectivity justifies the use of the tables of Laplace transforms since each Laplace transform corresponds to exactly one function. When we identify a transform in a table, we know the associated function is uniquely determined.

This uniqueness is also essential for the inversion problem, which asks whether we can recover a function from its Laplace transform.

Theorem 2.3.5 (Inversion Theorem for the Laplace Transform)**.** *Let $f : \mathbb{R}_+ \to \mathbb{R}$ be locally integrable and of exponential order $\alpha \in \mathbb{R}$. Let $F(s) = \mathcal{L}\{f(t)\}(s)$ be the Laplace transform of the function f, then for $t > 0$,*

$$f(t) = \frac{1}{2\pi \mathrm{i}} \int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty} F(s)e^{sx}\,ds \qquad \forall\, \Re(\sigma) > \alpha.$$

A proof of the Inversion Theorem for the Laplace Transform can be found in Earl.[VI] In practice, the Inversion Theorem for the Laplace Transform is evaluated using contour integration in the complex plane by closing the contour to the left and summing the residues of $F(s)e^{st}$ at the poles of $F(s)$.

2.4 Applications to Solving Initial Value Problems

The following examples demonstrate the transform-solve-invert procedure:

[V] L. Chierchia. *Injectivity of the Laplace Transform.* Lecture notes, Università degli Studi Roma Tre. 2019. URL: https://www.mat.uniroma3.it/users/chierchia/AM430_19_20/iniettivita-Laplace.pdf.

[VI] Earl, see n. IV.

Example 2.4.1 (First-Order Linear Initial Value Problem)**.** Consider the first order IVP

$$x'(t) + x(t) = 0, \qquad x(0) = 1.$$

Taking the Laplace transform of both sides and applying the derivative formula gives

$$sX(s) - 1 + X(s) = 0,$$

where $X(s) = \mathcal{L}\{x(t)\}(s)$. Rearranging,

$$X(s) = \frac{1}{s+1}.$$

From the table of Laplace transforms given in Table A.1, the Laplace transform of e^{-t} is equal to $\frac{1}{s+1}$. Therefore, using Injectivity of the Laplace Transform:

$$\mathcal{L}\{x(t)\}(s) = X(s) = \frac{1}{s+1} = \mathcal{L}\{e^{-t}\}(s) \quad \Longrightarrow \quad x(t) = e^{-t}.$$

Example 2.4.2 (Second-Order Initial Value Problem)**.** Consider the second-order IVP

$$x''(t) - 5x'(t) + 6x(t) = 0, \qquad x(0) = 2, \quad x'(0) = 2.$$

Taking the Laplace transform and substituting the initial conditions gives

$$\mathcal{L}\{x''\} - 5\mathcal{L}\{x'\} + 6X(s) = 0 \implies (s^2X(s) - 2s - 2) - 5(sX(s) - 2) + 6X(s) = 0,$$

so

$$X(s) = \frac{2s-8}{s^2-5s+6} = \frac{4}{s-2} - \frac{2}{s-3}.$$

Using the injectivity of the Laplace Transform:

$$x(t) = 4e^{2t} - 2e^{3t}.$$

2.5 Convolution and Integral Equations

Definition 2.5.1 (Convolution)**.** Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be locally integrable functions. The *convolution* of f and g is defined by

$$(f * g)(t) := \int_0^t f(t-\theta)g(\theta)\,d\theta, \qquad t \geq 0.$$

Note that the convolution operation is commutative, associative and distributive for any functions f, g, h and constants $\lambda, \mu \in \mathbb{C}$:

- $f * g = g * f$,
- $h * (f * g) = (h * f) * g$,
- $h * (\lambda f + \mu g) = \lambda(h * f) + \mu(h * g)$.

Theorem 2.5.1 (Convolution Theorem)**.** *Let $f, g : \mathbb{R}_+ \to \mathbb{R}$ be locally integrable functions of exponential orders $\alpha_f, \alpha_g \in \mathbb{R}$, respectively. Then for all $s \in \mathbb{C}$ with $\Re(s) > \max\{\alpha_f, \alpha_g\}$, the Laplace transform of the convolution satisfies*

$$\mathcal{L}\{(f * g)(t)\}(s) = \mathcal{L}\{f(t)\}(s)\,\mathcal{L}\{g(t)\}(s).$$

This means that the Laplace transform converts a convolution in the time domain into multiplication in the s-domain.

Proof. By definition,

$$\mathcal{L}\{(f * g)(t)\}(s) = \int_0^\infty \left(\int_0^t f(t-\theta)g(\theta)\, d\theta \right) e^{-st}\, dt.$$

The order of integration may be interchanged by Fubini's theorem for $\mathrm{Re}(s) > \max\{\alpha_f, \alpha_g\}$, yielding:

$$= \int_0^\infty \int_0^\infty f(t-\theta)g(\theta)e^{-st}\, dt\, d\theta.$$

Substitute $T = t - \theta$ (so $t = T + \theta$, $dt = dT$). The region of integration remains $T \geq 0$, $\theta \geq 0$, and the integral becomes

$$\int_0^\infty \int_0^\infty f(T)g(\theta)e^{-s(T+\theta)}\, dT\, d\theta = \left(\int_0^\infty f(T)e^{-sT}\, dT \right) \left(\int_0^\infty g(\theta)e^{-s\theta}\, d\theta \right) = \mathcal{L}\{f\}(s) \cdot \mathcal{L}\{g\}(s).$$

□

2.5.1 Application to an Integral Equation

Consider the following Volterra integral equation of the second kind, where $f(t)$ is a known kernel and $x(t)$ is the unknown function:

$$x(t) + \int_0^t f(t-\theta)x(\theta)\, d\theta = g(t),$$

with $g(t)$ a given forcing function. This can be expressed compactly using convolution as

$$x(t) + (f * x)(t) = g(t).$$

Taking the Laplace transform of both sides and applying the Convolution Theorem yields

$$X(s) + F(s)X(s) = G(s),$$

where $X(s) = \mathcal{L}\{x(t)\}$, $F(s) = \mathcal{L}\{f(t)\}$, and $G(s) = \mathcal{L}\{g(t)\}$.

Hence,

$$X(s) = \frac{G(s)}{1 + F(s)}.$$

The solution in the time domain is then obtained by applying the inverse Laplace transform

$$x(t) = \mathcal{L}^{-1}\left\{ \frac{G(s)}{1 + F(s)} \right\}$$

or using the injectivity and Table A.1. This shows how the convolution theorem reduces an integral equation to an algebraic equation in the transform domain.

Figure 2.2 shows an example of the application of the convolution integral. The top row shows two functions $f(\tau) = e^{-\tau}$ and $g(\tau) = e^{-2\tau}$ with their convolution result $(f * g)(t) = e^{-t} - e^{-2t}$. The bottom row shows the convolution for increasing values of t. For each fixed t, the blue curve shows $f(\tau)$, the red dashed curve shows $g(t-\tau)$, and the yellow shaded region represents the

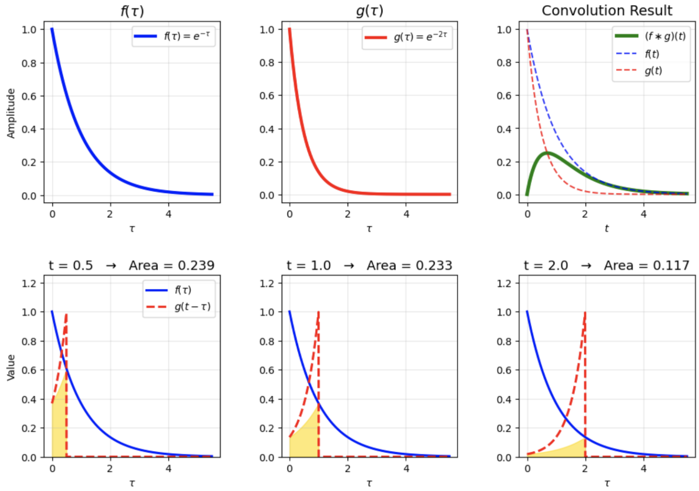

Figure 2.2: Graphical Interpretation of the Convolution Integral.

integrand $f(\tau)g(t-\tau)$. The area of this shaded region equals the value of the convolution at that particular t.

This is useful for understanding integral equations. Many Volterra-type integral equations can be expressed using convolutions. By replacing the convolution with its Laplace transform (which becomes $F(s)G(s)$), the integral equation is transformed into an algebraic equation that is easier to solve.

2.6 Applications of the Laplace Transform

Beyond integral equations, the same algebraic simplification extends to the analysis of linear dynamical systems through transfer functions.

2.6.1 Transfer Functions and the s-Domain Representation

Consider the higher order IVP with function $f(t)$ to be found and a known function $u(t)$

$$a_n f^{(n)}(t) + a_{n-1} f^{(n-1)}(t) + \cdots + a_1 f'(t) + a_0 f(t) = u(t),$$

where $a_k \in \mathbb{R}$ for $k = 0, 1, \ldots, n$ and $a_n \neq 0$, subject to the initial conditions $f^{(k)}(0) = 0$ for $k = 0, 1, \ldots, n-1$.

Taking the Laplace transform (bearing in mind the zero initial conditions) gives

$$\left(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0\right) F(s) = U(s).$$

Where $F(s) = \mathcal{L}\{f(t)\}(s)$ and $U(s) = \mathcal{L}\{u(t)\}(s)$. Hence, $F(s) = H(s)U(s)$, where

$$H(s) = \frac{1}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}.$$

This function is known as the *Transfer Function* of the system. The polynomial denominator determines the poles of $H(s)$, whose location in the complex plane determines the stability and the behaviour in the time domain. The solution is recovered by applying the inverse Laplace transform:

$$f(t) = \mathcal{L}^{-1}\{H(s)U(s)\}(t).$$

Transfer functions are used in control systems, electrical circuits, and signal processing. They allow differential equations to be analysed algebraically to study properties such as stability and system response. Since they are defined via the Laplace transform, injectivity ensures each transfer function uniquely determines the system's behaviour and that inversion is well-defined (Åström and Murray[VII]).

2.7 Conclusion

The Laplace transform is a method used to solve linear differential equations and Volterra integral equations. The key theoretical results – existence conditions, injectivity, and the inversion theorem, justify the standard "transform–solve–invert" procedure and guarantee that the recovered solution is unique.

Beyond differential equations, the convolution theorem extends this algebraic simplification to integral equations, while the transfer function $H(s)$ describes linear dynamical systems, allowing to analyse stability and frequency of the response. In each case, structure that is difficult to analyse in the time domain becomes tractable in the frequency domain

In real-life applications, the Laplace transform is widely used in control engineering (e.g. designing autopilot systems and PID controllers), electrical circuit analysis, signal processing, and mechanical vibration studies.

3 The Fourier Transform

The Fourier transform was named after the French mathematician Jean-Baptiste Joseph Fourier (1768–1830), who introduced the idea that any function could be represented as a sum of sinusoidal components while studying heat conduction. His ideas were later formalised and extended by mathematicians including Dirichlet and Riemann, establishing the Fourier transform as one of the most widely applied tools across mathematics, physics and engineering.

The Fourier transform extends the Fourier series to functions defined on the whole real line, replacing discrete frequencies with continuous ones. The Fourier transform allows a function to be analysed in either the time (spatial) domain or frequency domain. By converting differentiation into multiplication in the frequency domain, we can solve linear differential and partial differential equations.

3.1 Fourier Integrals and the Fourier Transform

3.1.1 Motivation from Fourier Series

The Fourier series of a periodic function $f : \mathbb{R} \to \mathbb{R}$ can be represented as the sum of complex exponentials:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

Frequencies (given by n) are discrete, reflecting the periodic nature of the function.

When we instead consider non-periodic functions defined on the entire real line $\mathbb{R}$, the discrete spectrum of the Fourier series becomes a continuous spectrum. The sum over discrete frequencies is replaced by an integral over a continuous frequency variable $\omega \in \mathbb{R}$, and the Fourier coefficients become a function $\hat{u}(\omega)$ of this continuous variable. This conversion from Fourier series to the Fourier integral is the generalisation when periodicity is removed.

[VII] K. J. Åström and R. M. Murray. "Transfer Functions". Unpublished lecture notes, California Institute of Technology. 2006. URL: https://www.cds.caltech.edu/~murray/books/AM05/pdf/am06-xferfcns_16Sep06.pdf.

3.1.2 Definition of the Fourier Transform

Definition 3.1.1 (The Fourier Transform)**.** Suppose that $u : \mathbb{R} \to \mathbb{R}$ (or $\mathbb{C}$) and $w : \mathbb{R} \to \mathbb{C}$ are bounded, continuous and such that:

$$\int_{-\infty}^{\infty} |u(t)|\, dt < \infty \qquad \text{and} \qquad \int_{-\infty}^{\infty} |w(\omega)|\, d\omega < \infty.$$

The Fourier transform of u is defined as

$$\mathcal{F}[u](\omega) = \int_{-\infty}^{\infty} e^{\mathrm{i}\omega t} u(t)\, dt, \qquad \omega \in \mathbb{R}.$$

The inverse Fourier transform of w is defined as

$$\mathcal{F}^{-1}[w](t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\mathrm{i}\omega t} w(\omega)\, d\omega, \qquad t \in \mathbb{R}.$$

These integrability and continuity assumptions ensure that the Fourier transform exists and that the inversion formula holds pointwise, i.e. $\mathcal{F}^{-1}[\mathcal{F}[u]](t) = u(t)$ for all $t \in \mathbb{R}$. The variable ω represents frequency, so the Fourier transform maps a function from the time domain to the frequency domain.

3.2 Key Properties

The Fourier transform converts multiplication in one domain into a shift (translation) in the other domain. This duality is captured by the two shift theorems.

Proposition 3.2.1 (First Shift Theorem)**.** For all $\omega_0 \in \mathbb{R}$, one has

$$\mathcal{F}\big[e^{-\mathrm{i}\omega_0 t} u(t)\big](\omega) = \mathcal{F}[u](\omega - \omega_0), \quad \omega \in \mathbb{R}.$$

In other words, multiplying by an (oscillatory) exponent in time domain is equivalent to a shift in frequency domain.

Proof.

$$\mathcal{F}\big[e^{-\mathrm{i}\omega_0 t} u(t)\big](\omega) = \int_{-\infty}^{\infty} e^{\mathrm{i}\omega t} e^{-\mathrm{i}\omega_0 t} u(t)\, dt = \int_{-\infty}^{\infty} e^{\mathrm{i}(\omega-\omega_0)t} u(t)\, dt = \mathcal{F}[u](\omega - \omega_0).$$

□

Proposition 3.2.2 (Second Shift Theorem)**.** Let $t_0 \in \mathbb{R}$ and define the time-shifted function $u_{t_0}(t) = u(t - t_0)$ for $t \in \mathbb{R}$. Then

$$\mathcal{F}[u_{t_0}](\omega) = e^{i\omega t_0} \mathcal{F}[u](\omega), \quad \forall\, \omega \in \mathbb{R}.$$

In other words, a shift in the time domain corresponds to multiplication by a complex exponential in the frequency domain.

Proof. For all $\omega \in \mathbb{R}$,

$$\mathcal{F}[u_{t_0}](\omega) = \int_{-\infty}^{\infty} e^{i\omega t} u(t - t_0)\, dt.$$

Make the substitution $s = t - t_0$ (so $t = s + t_0$ and $dt = ds$). The integral becomes

$$\int_{-\infty}^{\infty} e^{i\omega(s+t_0)} u(s)\, ds = e^{i\omega t_0} \int_{-\infty}^{\infty} e^{i\omega s} u(s)\, ds = e^{i\omega t_0} \mathcal{F}[u](\omega).$$

$\square$

Differentiation in one domain corresponds to multiplication by a linear factor in the other domain. This is one of the most useful properties of the Fourier transform.

Proposition 3.2.3 (Derivative Theorem)**.** If u is differentiable on $\mathbb{R}$, and $|u(t)| \to 0$ as $|t| \to \infty$, then

$$\mathcal{F}[u'](\omega) = -\mathrm{i}\omega \mathcal{F}[u](\omega), \quad \omega \in \mathbb{R}.$$

Proof. Integrating by parts, we obtain

$$\mathcal{F}[u'](\omega) = \int_{-\infty}^{\infty} e^{\mathrm{i}\omega t} u'(t)\, dt = \left[e^{\mathrm{i}\omega t} u(t) \right]_{-\infty}^{\infty} - \mathrm{i}\omega \int_{-\infty}^{\infty} e^{\mathrm{i}\omega t} u(t)\, dt = -\mathrm{i}\omega \int_{-\infty}^{\infty} e^{\mathrm{i}\omega t} u(t)\, dt = -\mathrm{i}\omega \mathcal{F}[u](\omega),$$

as required. $\square$

Proposition 3.2.4 (Reverse Derivative Theorem)**.** If u is differentiable on $\mathbb{R}$, and $|u(\omega)| \to 0$ as $|\omega| \to \infty$, one has

$$\mathcal{F}^{-1}[u'](t) = -it\, \mathcal{F}^{-1}[u](t), \quad t \in \mathbb{R}.$$

Proof. By definition of the inverse Fourier transform,

$$\mathcal{F}^{-1}[u'](t) = (2\pi)^{-1} \mathcal{F}[u'](-t).$$

The Fourier transform of the derivative satisfies

$$\mathcal{F}[u'](\omega) = i\omega\, \mathcal{F}[u](\omega)$$

for all $\omega \in \mathbb{R}$. Substituting $\omega = -t$ gives

$$\mathcal{F}[u'](-t) = i(-t)\, \mathcal{F}[u](-t) = -it\, \mathcal{F}[u](-t).$$

Therefore,

$$\mathcal{F}^{-1}[u'](t) = (2\pi)^{-1} \big(-it\, \mathcal{F}[u](-t) \big) = -it \cdot (2\pi)^{-1} \mathcal{F}[u](-t).$$

By definition of the inverse Fourier transform again,

$$(2\pi)^{-1} \mathcal{F}[u](-t) = \mathcal{F}^{-1}[u](t),$$

and hence

$$\mathcal{F}^{-1}[u'](t) = -it\, \mathcal{F}^{-1}[u](t),$$

as claimed. $\square$

Theorem 3.2.1 (Convolution Theorem)**.** *If the functions f, g and $f * g$ have well-defined Fourier transforms, then*

$$\mathcal{F}[f * g] = \mathcal{F}[f] \mathcal{F}[g].$$

In other words, convolution in the time domain becomes pointwise multiplication in the frequency domain.

The proof is essentially identical in structure to the corresponding proof for the Laplace transform. Substitute the definition of the convolution

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\, d\tau$$

into the Fourier integral and change the order of integration.

3.3 Applications to Partial Differential Equations

The Fourier transform provides a method for solving linear partial differential equations (PDEs) on the real line $x \in \mathbb{R}$. The key idea is that differentiation in physical space becomes multiplication in frequency space, reducing a PDE to an ODE.

3.3.1 General Strategy

Given some PDE in $u(x,t)$ (and denoting the Fourier transform of u as $\widehat{u}$):

1. Take the Fourier transform with respect to x:

$$\widehat{u}(\omega, t) = \mathcal{F}[u(x,t)](\omega) = \int_{-\infty}^{\infty} u(x,t)e^{\mathrm{i}\omega x} dx.$$

2. Use the differentiation properties:

$$\mathcal{F}[u_x] = -\mathrm{i}\omega\widehat{u}, \qquad \mathcal{F}[u_{xx}] = -\omega^2\widehat{u},$$

   to convert spatial derivatives into algebraic factors in ω.

3. Solve the resulting ODE in t for each fixed ω.
4. Apply the inverse Fourier transform to recover $u(x,t)$.

3.3.2 The Heat Equation on $\mathbb{R}$

Consider the heat equation with thermal diffusivity $\kappa > 0$:

$$\begin{cases} u_t = \kappa u_{xx}, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = h(x). \end{cases}$$

Taking the Fourier transform with respect to the spatial variable x and denoting $\widehat{u}(\omega,t) := \mathcal{F}[u(\cdot,t)](\omega)$, we obtain the transformed system

$$\begin{cases} \frac{\partial \widehat{u}}{\partial t}(\omega,t) = -\kappa\omega^2\widehat{u}(\omega,t), & \omega \in \mathbb{R},\ t \geq 0, \\ \widehat{u}(\omega,0) = \mathcal{F}[h](\omega). \end{cases}$$

This is now a simple ODE in t for each fixed ω. Its solution is

$$\widehat{u}(\omega,t) = \mathcal{F}[h](\omega)\, e^{-\kappa\omega^2 t}.$$

Recognising that $e^{-\kappa\omega^2 t} = \mathcal{F}[\Phi(\cdot,t)](\omega)$, where Φ is the fundamental solution (heat kernel)

$$\Phi(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{x^2}{4\kappa t}\right),$$

and applying the Convolution Theorem, we recover the solution in physical space:

$$u(x,t) = (\Phi(\cdot,t) * h)(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-z)^2}{4\kappa t}\right) h(z)\, dz.$$

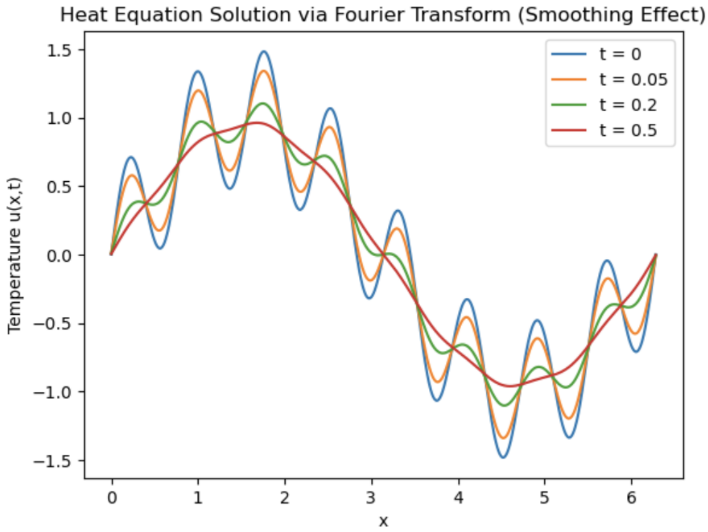

Figure 3.1: Solution of the heat equation at successive times t.

The factor $e^{-\kappa\omega^2 t}$ exponentially dampens high-frequency components, which explains why the solution becomes smoother as t increases.

Figure 3.1 shows $u(x,t)$ at successive times t in the interval $x \in [0,6]$ with thermal diffusivity $\kappa = 0.1$. At $t = 0$, the curve is the initial condition $h(x)$; as t increases the solution flattens and spreads, reflecting heat diffusing from hot to cold regions. The exponential part $e^{-\kappa\omega^2 t}$ damps high-frequency components faster than the low-frequency ones, so only slow and smooth variations survive at large t.

The Fourier transform provides an elegant method for solving the heat equation on the infinite line. By transforming the PDE into the frequency domain, the solution becomes a simple multiplication by the damping factor $e^{-\kappa\omega^2 t}$. Applying the inverse transform yields the Gaussian convolution form.

3.3.3 Laplace's Equation

The Fourier transform can also be applied to elliptic equations on bounded domains. Consider Laplace's equation on the square $[0,L] \times [0,L]$:

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < L,\ 0 < y < L, \\ u(0,y) = 0, \quad u(L,y) = 0, \quad u(x,L) = 0, & 0 \le y \le L, \\ u(x,0) = f(x), & 0 \le x \le L. \end{cases}$$

To solve this, we use separation of variables. Suppose that the solution can be written as a product of the function X and Y where X is a function of x only and Y is a function of y only, namely $u(x,y) = X(x)Y(y)$. Substituting into Laplace's equation yields

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda.$$

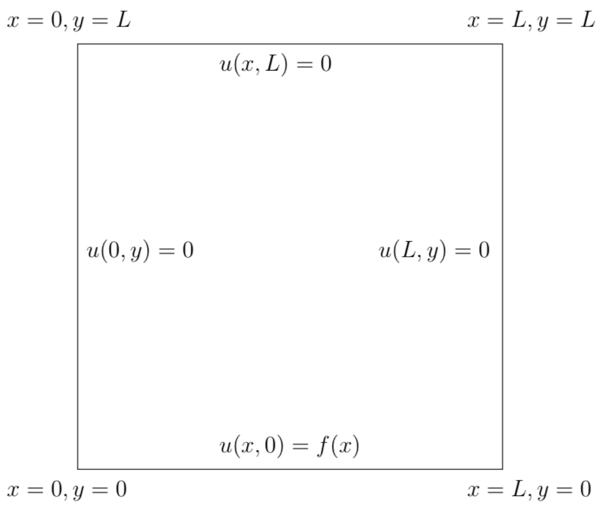

Figure 3.2: Dirichlet boundary conditions for Laplace's equation on the square $[0, L] \times [0, L]$.

From the homogeneous boundary conditions in x ($X(0) = X(L) = 0$), we obtain the eigenvalue problem

$$X'' + \lambda X = 0, \quad X(0) = X(L) = 0,$$

with solutions

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \ldots$$

For each n, the corresponding Y-equation is

$$Y'' - \lambda_n Y = 0 \implies Y_n(y) = A_n \cosh\left(\frac{n\pi y}{L}\right) + B_n \sinh\left(\frac{n\pi y}{L}\right).$$

The boundary condition $u(x, L) = 0$ implies $Y_n(L) = 0$, which forces

$$Y_n(y) = C_n \sinh\left(\frac{n\pi(L - y)}{L}\right).$$

The general solution is therefore the superposition

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi(L - y)}{L}\right).$$

Applying the remaining boundary condition $u(x, 0) = f(x)$ gives the Fourier sine series coefficients

$$B_n \sinh\left(\frac{n\pi L}{L}\right) = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx,$$

so

$$u(x, y) = \sum_{n=1}^{\infty} b_n \frac{\sinh\left(\frac{n\pi(L-y)}{L}\right)}{\sinh(n\pi)} \sin\left(\frac{n\pi x}{L}\right),$$

where b_n are the Fourier sine coefficients of $f(x)$.

3.4 Conclusion

The Fourier transform provides a method for solving PDEs on unbounded domains by converting differential equations into simple algebraic equations in the frequency domain. This simplifies analysis and gives solutions expressed as convolutions with fundamental kernels, such as the Gaussian heat kernel. The examples illustrate how the transform reveals both the structure and qualitative behaviour of solutions.

The Fourier transform is used in signal and image processing (e.g. JPEG compression and noise reduction), medical imaging (MRI and CT reconstruction), audio engineering, and geophysics. It is also essential for modelling heat diffusion, pollutant transport, and wave propagation in engineering applications.

4 The Mellin Transform

In contrast to the Fourier and Laplace transforms, which were originally developed to address physical problems, the Mellin transform arose from a more theoretical context. Its earliest appearance can be traced to the work of Bernhard Riemann, who applied it in the study of the Riemann zeta function, and was later developed systematically by the Finnish mathematician Hjalmar Mellin, after whom it is now named (Ovarlez[I]).

The Mellin transform is particularly well-suited to problems involving scaling and multiplicative structure. It is especially effective for solving certain boundary value problems and for analysing differential equations that contain the Euler operator $x\frac{d}{dx}$ (which frequently appears in electrical engineering, fluid mechanics, and problems with power-law behaviour). In this sense, the Mellin transform provides a method analogous to the Laplace transform, but adapted to multiplicative (scaling) problems rather than additive (translational) ones: it converts differential equations involving $x\frac{d}{dx}$ into algebraic equations in the transform variable s.

4.1 Basic Properties

Definition 4.1.1 (Mellin Transform)**.** The Mellin transform is the operation mapping the function $f : \mathbb{R}_+ \to \mathbb{C}$ to the function F and is defined by

$$F(s) = \mathcal{M}\{f(t)\}(s) = \int_0^\infty f(t)\, t^{s-1}\, dt, \tag{4.1}$$

where $s = \sigma + \mathrm{i}\tau$ is the complex frequency with $\sigma, \tau \in \mathbb{R}$. The complex variable s plays a role analogous to the frequency variable in the Fourier transform.

The real part $\sigma = \Re(s)$ controls the growth or decay of the function through the factor $t^{\sigma-1}$, while the imaginary part $\tau = \Im(s)$ introduces oscillatory behaviour via the factor $t^{\mathrm{i}\tau} = e^{\mathrm{i}\tau \log t}$.

The Mellin transform thus analyses the function f in terms of multiplicative (scaling) rather than additive (frequency) structure.

Theorem 4.1.1 (Conditions for Existence)**.** *The Mellin transform exists for all values of $s \in \mathbb{C}$ for which*

$$\int_0^\infty |f(t)|\, t^{\sigma-1}\, dt < \infty,$$

where $\sigma = \Re(s)$.

In particular, if

$$\int_0^\infty |f(t)|\, t^{c-1}\, dt < \infty$$

[I]Jean-Philippe Ovarlez. *The Mellin Transform.* URL: https://www.jeanphilippeovarlez.com/publications/ewExternalFiles/the%20mellin%20.pdf.

for some constant $c \in \mathbb{R}$. The Mellin transform exists and is absolutely convergent for all $s = c + \mathrm{i}\tau$.

The Mellin transform is sensitive to a function's asymptotic behaviour as $t \to 0^+$ and as $t \to +\infty$, due to its power-law kernel t^{s-1}. This leads to a restriction on the values of s for which the transform converges, known as the region of convergence (RoC).

Unlike the Laplace transform, which typically converges in a half-plane, the Mellin transform generally converges in a vertical strip in the complex plane. This reflects the need to control the behaviour of the integrand at both ends of the domain.

The Region of Convergence (RoC) is typically an open vertical strip of the form

$$\{s \in \mathbb{C} \mid a < \mathrm{Re}(s) < b\},$$

called the fundamental strip, where the constants a and b (not necessarily finite) depend on the behaviour of $f(t)$ at $t = 0^+$ and as $t \to \infty$. More precisely:

As $t \to 0^+$: if $f(t) \sim t^{-\gamma}$, then the integral converges for $\mathrm{Re}(s) > a$, where $a = \gamma$. As $t \to \infty$: if $f(t)$ grows or decays like t^{δ}, then the integral converges for $\mathrm{Re}(s) < b$, where $b = \delta$.

Hence, the fundamental strip is determined by the balance $a < \mathrm{Re}(s) < b$.

Example 4.1.1. Consider $f(x) = e^{-x}$. As $x \to 0^+$, $f(x) \sim 1$, so we require $\Re(s) > 0$. As $x \to \infty$, the exponential decay dominates any power, so there is no upper restriction. Hence, the Mellin transform converges for: $\Re(s) > 0$.

4.2 Key Properties of the Mellin Transform

Theorem 4.2.1 (Injectivity of the Mellin Transform)**.** *Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be functions whose Mellin transforms $\mathcal{M}\{f\}(s)$ and $\mathcal{M}\{g\}(s)$ exist in a common vertical strip of the complex plane. If*

$$\mathcal{M}\{f\}(s) = \mathcal{M}\{g\}(s)$$

for all s in that strip, then

$$f(x) = g(x)$$

almost everywhere on $(0, \infty)$.

"Almost everywhere" reflects that the Mellin transform is an integral operator. Two functions may differ at a single point, finitely many points, or on a countable set, and still have the same Mellin transform (Paris and Kaminski[II]).

Proof. Let $h(x) = f(x) - g(x)$. Suppose the Mellin transforms agree on a common vertical line $\Re(s) = c$ inside the strip of convergence. Then

$$\mathcal{M}\{h\}(s) = 0 \quad \text{for all } s \text{ with } \Re(s) = c,$$

that is,

$$\int_0^\infty h(x) x^{s-1}\, dx = 0.$$

Write $s = c + \mathrm{i}\omega$ with $\omega \in \mathbb{R}$. The equation becomes

$$\int_0^\infty h(x) x^{c-1} x^{\mathrm{i}\omega}\, dx = 0.$$

[II] R. B. Paris and D. Kaminski. *Asymptotics and Mellin-Barnes Integrals.* Cambridge University Press, 2001.

Make the substitution $x = e^t$, so $dx = e^t\,dt$ and t runs from $-\infty$ to $+\infty$. This yields

$$\int_{-\infty}^{\infty} h(e^t)e^{ct}e^{i\omega t}\,dt = 0.$$

Define $H(t) = h(e^t)e^{ct}$. Then

$$\int_{-\infty}^{\infty} H(t)e^{i\omega t}\,dt = 0 \quad \text{for all } \omega \in \mathbb{R}.$$

This means that the Fourier transform of H is identically zero. Since the Fourier transform is injective (under the integrability conditions guaranteed by the assumptions on f and g), it follows that $H(t) = 0$ almost everywhere. Therefore $h(e^t)e^{ct} = 0$ almost everywhere, which implies $h(x) = 0$ almost everywhere on $(0, \infty)$. Hence $f(x) = g(x)$ almost everywhere. This establishes the injectivity of the Mellin transform via its correspondence with the Fourier transform under the logarithmic change of variables $x = e^t$. □

An immediate consequence of injectivity is that it justifies the use of the tables of Mellin transforms (just like Laplace transforms). Since each Mellin transform corresponds to exactly one function, these tables can be used reliably. When we identify a transform in a table, we know the associated function is uniquely determined.

Theorem 4.2.2 (Scaling Property)**.** *Let $a > 0$. Then*

$$\mathcal{M}\{f(at)\}(s) = a^{-s}\mathcal{M}\{f(t)\}(s).$$

Proof. By definition,

$$\mathcal{M}\{f(at)\}(s) = \int_0^{\infty} f(at)t^{s-1}\mathrm{d}t.$$

Let $u = at$, so that $t = u/a$ and $\mathrm{d}t = \mathrm{d}u/a$. Then

$$\mathcal{M}\{f(at)\}(s) = \int_0^{\infty} f(u)\left(\frac{u}{a}\right)^{s-1}\frac{\mathrm{d}u}{a} = a^{-s}\int_0^{\infty} f(u)u^{s-1}\mathrm{d}u = a^{-s}\mathcal{M}\{f(t)\}(s).$$

□

Note that the scaling property requires $a > 0$, since the Mellin transform is defined only for positive real arguments. For negative scaling one would first need to extend the function f to negative values.

Scaling the input of a function by a constant $a > 0$ results in the Mellin transform being multiplied by a^{-s}. This is analogous to the Fourier transform, where translating a function by a constant multiplies its transform by a phase factor. In the Mellin case, scaling leads to multiplication by a power a, making it suitable for problems involving power-laws or scaling.

Theorem 4.2.3 (Multiplication by a Power)**.** *Let $\alpha \in \mathbb{C}$. Then*

$$\mathcal{M}\{t^{\alpha}f(t)\}(s) = \mathcal{M}\{f(t)\}(s+\alpha).$$

Proof.

$$\mathcal{M}\{t^{\alpha}f(t)\}(s) = \int_0^{\infty} t^{\alpha}f(t)t^{s-1}dt = \int_0^{\infty} f(t)t^{(s+\alpha)-1}dt, = \mathcal{M}\{f(t)\}(s+\alpha).$$

□

This means that multiplication by a power of t produces a shift in the transform variable.

4.2.1 Relation to the Laplace and Fourier Transforms

The Mellin transform is related to both the Laplace and Fourier transforms through a logarithmic change of variables (Forster[III]). The substitution $x = e^{-t}$ converts the multiplicative scaling in x into a translation in t, which is why the Mellin transform behaves like a Fourier or Laplace transform once viewed on a logarithmic scale.

Recall the Mellin transform:

$$\mathcal{M}\{f(x)\}(s) = \int_0^{\infty} f(x)\, x^{s-1}\, dx. \tag{4.2}$$

Consider the change of variables $x = e^{-t}$ (so that $dx = -e^{-t}dt$), then the integral becomes

$$\mathcal{M}\{f(x)\}(s) = \int_{-\infty}^{\infty} f(e^{-t})\, e^{-st}\, dt. \tag{4.3}$$

Let $g(t) = f(e^{-t})$, we obtain

$$\mathcal{M}\{f(x)\}(s) = \int_{-\infty}^{\infty} g(t)\, e^{-st}\, dt, \tag{4.4}$$

which is the *two-sided Laplace transform*[IV] of $g(t)$ (Jacqueline Bertrand[V]). Which is why the Mellin transform may be regarded as the multiplicative analogue of the Laplace transform.

The connection with the Fourier transform follows by writing $s = \sigma + \mathrm{i}\tau$ within the strip of convergence:

$$\mathcal{M}\{f\}(\sigma + \mathrm{i}\tau) = \int_{-\infty}^{\infty} g(t)\, e^{-\sigma t} e^{-\mathrm{i}\tau t}\, dt = \mathcal{F}\{g(t)e^{-\sigma t}\}(\tau). \tag{4.5}$$

Thus, along vertical lines $\Re(s) = \sigma$, the Mellin transform reduces to a Fourier transform of $f(e^{-t})$ with exponential damping. Equivalently, it may be interpreted as a Fourier transform on the logarithmic scale $t = -\log x$.

4.3 Conclusion

The Mellin transform is suited to functions with scaling and power-law behaviour. Its properties mirror those of the Laplace transform and the substitution $x = e^{-t}$ formalises this connection, showing the Mellin transform is equivalent to a bilateral Laplace transform, and along vertical lines to a Fourier transform. As a result, problems involving multiplicative structure that are difficult to analyse directly become algebraically manageable in the Mellin domain.

Some applications of the Mellin transform are asymptotic analysis of algorithms, fractal geometry and option pricing in mathematical finance (via Mellin-Barnes integrals, Yoon[VI]). It is also used in engineering for modelling power-law phenomena such as turbulence and network traffic.

[III] Otto Forster. *Analytic Number Theory: Lecture Notes (Chapter 12: Laplace and Mellin Transform)*. Accessed: 31 March 2026. 2001. URL: https://www.mathematik.uni-muenchen.de/~forster/v/ann/annth_12.pdf.

[IV] The two-sided Laplace transform integrates over the entire real line $(-\infty, \infty)$ rather than just $[0, \infty)$. It arises naturally here because the substitution $x = e^{-t}$ maps $\mathbb{R}_+$ to all of $\mathbb{R}$.

[V] Jean-Philippe Ovarlez Jacqueline Bertrand Pierre Bertrand. *Integral Transform Research Paper*. 2021. URL: https://hal.science/hal-03152634/document.

[VI] Ji-Hun Yoon. *Mellin Transform Method for European Option Pricing with Hull-White Stochastic Interest Rate*. 2024. URL: https://projecteuclid.org/journals/journal-of-applied-mathematics/volume-2014/issue-none/Mellin-Transform-Method-for-European-Option-Pricing-with-Hull-White/10.1155/2014/759562.full?tab=ArticleFirstPage.

5 The Weierstrass Transform

The **Weierstrass transform** (also called the Gauss-Weierstrass transform) is a mathematical tool that smooths out functions. It comes from the study of the *heat equation*, which describes how heat spreads and evens out over time. Applying the Weierstrass transform to a function is mathematically the same as letting heat diffuse for a short time, starting from that function as the initial temperature distribution.

The transform works by averaging nearby values of the original function, with closer points having much more influence (as the average weights follow a bell-shaped *Gaussian* curve). This process removes sharp changes and noise, producing a smooth result.

Because of this smoothing property, the Weierstrass transform is used in:

- approximation theory (proving polynomials get arbitrarily close to continuous functions)
- signal and image processing (removing noise while keeping important features)
- studying solutions of the heat equation and other diffusion processes.

5.1 Definition

Definition 5.1.1 (The Weierstrass transform)**.** Let $f : \mathbb{R} \to \mathbb{R}$. Then, the Weierstrass transform of f, denoted $F = W[f]$, is given by

$$F(x) = W[f](x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(y)\, e^{-(x-y)^2/4}\, dy. \tag{5.1}$$

This integral can also be written in the shifted form:

$$F(x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(x-y)\, e^{-y^2/4}\, dy. \tag{5.2}$$

The transform is well-defined whenever f is (locally) integrable or belongs to a suitable larger class (e.g., bounded continuous functions), but it may fail to exist for functions that grow too fast at infinity. For example, $f(x) = e^{x^2}$ grows too rapidly for the integral to converge.

A more detailed definition and equivalent representations of the Weierstrass transform can be found in Rooney[I] .

5.2 General Properties

The Weierstrass transform has several important and useful properties:

Theorem 5.2.1 (Translation Invariance (Shift Property))**.** *If $g(x) = f(x + a)$, then*

$$W[g](x) = W[f](x + a).$$

That is, if the input function is shifted by a constant a, the output is shifted by the same amount.

[I] P. G. Rooney. *A Generalization of an Inversion Formula for the Gauss Transformation.* 2018. URL: https://www.cambridge.org/core/journals/canadian-mathematical-bulletin/article/generalization-of-an-inversion-formula-for-the-gauss-transformation/64991CE828124FE74661F1844708A9C1.

Proof. By definition,

$$W[g](x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} g(y)\, e^{-(x-y)^2}\, dy.$$

Substitute $g(y) = f(y+a)$:

$$W[g](x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(y+a)\, e^{-(x-y)^2}\, dy.$$

Make the change of variable $z = y + a$, so $dy = dz$. Then

$$W[g](x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(z)\, e^{-(x-(z-a))^2}\, dz = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(z)\, e^{-((x+a)-z)^2}\, dz.$$

The right-hand side is precisely $W[f](x+a)$. Therefore,

$$W[g](x) = W[f](x+a).$$

□

Theorem 5.2.2 (The Inverse Transform)**.** *The Weierstrass transform is invertible for suitable*[II] *classes of functions, meaning that the original function can be recovered from its transform.*

Unlike the Fourier and Laplace transforms, there is no simple general inversion formula in closed form. We invert the transform by reversing its smoothing effect, rather than through an explicit formula. A generalisation of the Inversion and proof can be found in Rooney[III] *.*

5.3 Connection to the Heat Equation

As derived in Section 3.3.2 using the Fourier transform, the solution to the heat equation

$$u_t = \kappa u_{xx}, \qquad x \in \mathbb{R},\ t > 0,$$

with initial condition $u(x,0) = h(x)$, is given by its convolution with the Gaussian heat kernel

$$u(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \int_{-\infty}^{\infty} h(t) \exp\left(-\frac{(x-t)^2}{4\kappa t}\right) dt.$$

This shows that the time-evolution operator for the heat equation is convolution with a Gaussian whose variance grows linearly with time. Compared with the definition of the Weierstrass transform,

$$W[f](x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(y)\, e^{-(x-y)^2/4}\, dy,$$

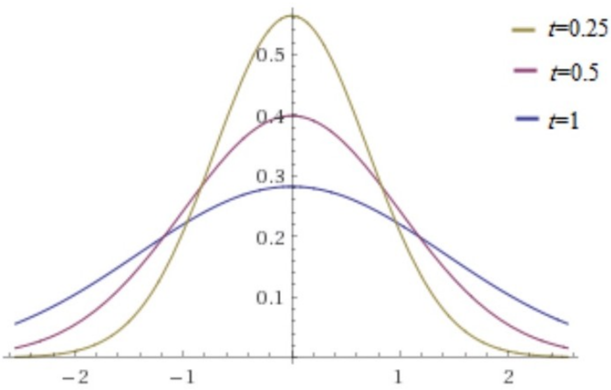

Figure 5.1: Plots of the heat kernel in one space and one time dimension, drawn at successive times. The curve is a Gaussian whose height increases without bound as $t \to 0+$.

we see that $W[f]$ is precisely the solution of the heat equation at time $t = 1$ when the diffusion constant is set to $\kappa = 1/4$. The normalisation factor is chosen so the heat kernel integrates to 1

[II] "Suitable" typically refers to functions that are sufficiently well-behaved, for example, functions that are continuous and do not grow too rapidly at infinity, so that the transform is well-defined.

[III] Rooney, see n. I.

for every $t > 0$,

$$\int_{-\infty}^{\infty} K(x,t)\,dx = 1,$$

thereby conserving the total heat (illustrated in Figure 5.1, Skinner[IV]).

As $t \to 0^+$, the heat kernel $K(x,t)$ becomes taller and narrower while preserving unit area, and converges to the Dirac delta function $\delta(t)$. This is why the solution $u(x,t)$ recovers the initial condition $u(x,0) = h(x)$ in the limit as $t \to 0^+$. Note that t here represents forward time evolution (diffusion forward in time), not going backwards.

The Weierstrass transform is the heat semigroup operator. It describes how the initial function f is smoothed and diffused under the heat equation.

The Fourier method reveals why this smoothing occurs: taking the Fourier transform of the heat equation shows that each frequency component of the initial data, $\widehat{h}(\omega) = \mathcal{F}[h](\omega)$, is multiplied by the factor $e^{-\kappa\omega^2 t}$. Since the exponent becomes more negative for larger $|\omega|$, higher spatial frequencies (rapid oscillations) decay much faster than lower frequencies. This frequency-dependent damping is the mathematical reason the Weierstrass transform acts as a powerful smoothing operator in analysis and in applications such as signal and image processing. (The idea of using the Weierstrass Transform with the heat equation can be seen in Lototsky[V] .)

5.4 The Weierstrass Transform as a Smoothing Operator

The Weierstrass transform shows how an integral transform can act as a **smoothing operator**, suppressing high-frequency components, as seen from the Fourier multiplier $e^{-\kappa\omega^2 t}$.

The noisy function in Figure 5.2 exhibits oscillations and spikes, corresponding to high-frequency components. Applying the Weierstrass transform attenuates these, producing a smoother function that more closely reflects the true signal.

The noisy signal (blue) is smoothed using Gaussian convolution, producing a curve (orange) that approximates the true signal (green dashed). Rapid oscillations are damped while the broad shape is preserved, because the Fourier multiplier $e^{-\omega^2}$ suppresses components with large $|\omega|$.

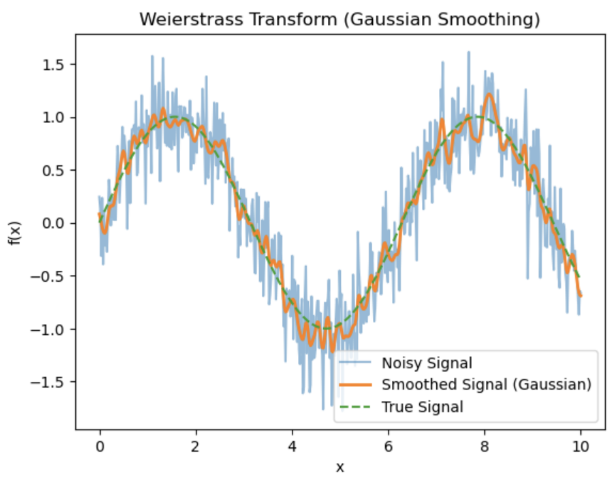

Figure 5.2: The Weierstrass transform as a smoothing operator.

5.5 Conclusion

The Weierstrass transform smooths functions by convolving them with a Gaussian kernel and is equivalent to evolving an initial temperature distribution under the heat equation for one

[IV] D. B. Skinner. "The Heat Equation". Lecture notes for Part IB Mathematical Methods, Department of Applied Mathematics and Theoretical Physics (DAMTP), University of Cambridge. 2015. URL: https://www.damtp.cam.ac.uk/user/dbs26/1BMethods/Heat.pdf.

[V] Sergey Lototsky. *The Weierstrass Approximation Theorem*. Lecture notes, University of Southern California (USC). Nov. 2022. URL: https://dornsife.usc.edu/sergey-lototsky/wp-content/uploads/sites/211/2023/06/WeierApprTh.pdf (visited on 04/13/2026).

unit of time. In the Fourier domain, this smoothing has a precise explanation; each frequency component is damped by $e^{-\kappa\omega^2 t}$, with higher frequencies suppressed more strongly. This makes the Weierstrass transform a tool in approximation theory, signal processing, and the study of diffusion processes.

6 Choosing the Appropriate Transform

The choice of integral transform depends on several key factors: the domain of the problem, the kernel of the transform, its action on differential operators, and the nature of the underlying physical or mathematical structure. These characteristics are summarised for the Laplace, Fourier, Mellin, and Weierstrass transforms in Table 6.1.

Table 6.1: Comparison of Transforms

Transform	**Domain**	**Kernel**	**Effect on derivatives**	**Best for**
Laplace	$t \geq 0$	e^{-st} $(s = \sigma + \mathrm{i}\tau)$	$f^{(n)}(t) \to s^n F(s)$ (with initial terms)	Initial value problems, linear ODEs, control systems
Fourier	$t \in \mathbb{R}$	$e^{i\omega t}$	$f'(t) \to i\omega \hat{f}(\omega)$	Frequency analysis, PDEs on $\mathbb{R}$
Mellin	$t > 0$	t^{s-1} $(s = \sigma + \mathrm{i}\tau)$	Scaling corresponds to shifts in s	Scaling laws, asymptotics, special functions
Weierstrass	$x \in \mathbb{R}$	$\frac{1}{\sqrt{4\pi}} e^{-(x-y)^2/4}$	Differentiation commutes with transform	Smoothing, heat/diffusion equation

The choice of transform depends on the problem's domain, the type of operation (derivative, scaling, or smoothing), initial/boundary conditions, and convergence requirements.

6.1 Laplace vs. Fourier: Applications to IVPs and BVPs

6.1.1 Difference between IVPs and BVPs

IVPs and BVPs arise in different contexts and exhibit different behaviours. An IVP specifies all conditions at a single point, typically representing the starting state of a system evolving in time and a BVP imposes conditions at multiple points, usually at the boundaries of a spatial domain, and is associated with equilibria or steady-state.

6.1.2 Strengths and Typical Uses

Table 6.2: Strengths and Typical Uses

	Laplace Transform	**Fourier Transform/Series**
Strengths	Handles causality, transients, unstable modes; initial conditions built-in	Frequency decomposition; excellent for unbounded/periodic domains
Typical uses	IVPs (time-dependent with ICs at $t = 0$); linear ODEs/PDEs; control systems	BVPs; whole-line PDEs; steady-state; frequency analysis

For initial-boundary-value problems (IBVPs), a hybrid approach is often highly effective: one applies the Laplace transform in time (to handle initial conditions) and the Fourier transform (or Fourier sine/cosine transform) in space (to handle boundary conditions).

A classic example is the heat equation on the semi-infinite line:

$$u_t = \kappa u_{xx}, \quad x > 0,\ t > 0,$$

subject to the initial condition $u(x, 0) = f(x)$ and the boundary condition $u(0, t) = 0$. Taking the Laplace transform in t and the Fourier sine transform in x reduces the IBVP to an algebraic equation that can be solved explicitly.

This combination exploits the strengths of both transforms and is a standard technique for solving linear PDEs on unbounded or semi-bounded spatial domains with time-dependent behaviour.

6.2 Laplace vs. Mellin: Time-Evolution vs Scaling and Power Laws

The main distinction between the Laplace and Mellin transforms lies in the geometry of the problems they are designed to solve. The Laplace transform is naturally suited to translation-invariant systems and initial-value problems, while the Mellin transform excels in scale-invariant systems involving power laws and multiplicative structures.

Table 6.3: Comparison of the Laplace and Mellin Transforms

Feature	**Laplace Transform**	**Mellin Transform**
Domain	$t \in [0, \infty)$ (linear scale)	$x \in (0, \infty)$ (logarithmic scale)
Kernel	Exponential e^{-st}	Power-law x^{s-1}
Primary Strength	Converts additive shifts $f(t-a)$ into phase factors e^{-as}	Converts multiplicative scaling $f(ax)$ into power factors a^{-s}
Convergence	Right half-plane $\Re(s) > \sigma$ (generally stable due to exponential decay)	Vertical strip $a < \Re(s) < b$ (often more restricted)
Weakness	Struggles with variable-coefficient equations (e.g. Euler-type)	Cannot handle exponential growth (e.g. e^t)
Best Used For	Linear ODEs/PDEs with constant coefficients and initial conditions at $t = 0$	Scaling problems, Cauchy–Euler equations, asymptotic analysis, and fractal-related structures

6.3 Fourier vs. Weierstrass: Frequency vs. Smoothing

The difference between the Fourier and Weierstrass transforms is in their kernel. The Fourier transform uses an oscillatory kernel $e^{-\mathrm{i}\omega t}$, whereas the Weierstrass transform uses a Gaussian kernel $e^{-\frac{(x-y)^2}{4}}$. The change from complex oscillation to real-values decay changes the transform.

The choice between the Fourier and Weierstrass transforms depends on whether the goal is to preserve frequency information or to smooth and regularise the function.

The Fourier transform is used to analyse frequencies of a function. Its kernel, namely $e^{-\mathrm{i}\omega t}$, is oscillatory and has constant magnitude.

The unit-magnitude means the Fourier kernel does not decay, dampen, or smooth the signal. Instead, it re-expresses the original function in terms of its constituent frequencies, preserving all spectral information. This means the Fourier transform is used when we have periodic structure, oscillatory behaviour, or hidden frequency components.

The Weierstrass transform acts as a smoothing operator, meaning it replaces each point with a local weighted average using a Gaussian kernel. The transform is defined by

Table 6.4: Comparison of the Fourier and Weierstrass Transforms

Feature	**Fourier Transform**	**Weierstrass Transform**
Kernel Type	Complex oscillatory kernel: $e^{-\mathrm{i}\omega x}$	Gaussian kernel: $\frac{1}{\sqrt{4\pi t}}e^{-\frac{(x-y)^2}{4t}}$
Spectral Effect	Decomposes a function into its discrete or continuous frequency components	Acts as a low-pass filter by damping high-frequency components
Structural Goal	Identifies hidden periodicities, oscillatory structure, and frequency content	Smooths irregularities, noise, and discontinuities through Gaussian averaging
Operator Role	Best suited to wave-type operators and oscillatory equations. For example: $u_{tt} = c^2 u_{xx}$	Best suited to diffusion and heat-type operators. For example: $u_t = k u_{xx}$

$$W[f](x) = \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} f(y) e^{\frac{-(x-y)^2}{4}} dy,$$

The kernel $e^{-(x-y)^2}$ decays rapidly as $|x-y|$ increases. The Fourier kernel oscillates indefinitely, whereas the Gaussian kernel is localised – nearby values contribute strongly, while distant values contribute little. This smooths sharp transitions, reducing noise, and eliminating small-scale irregularities.

7 Conclusion

This project summarised four integral transforms – Laplace, Fourier, Mellin and Weierstrass. Each transform is adapted to a specific class of mathematical problems based on its kernel and domain.

A comparative summary of the different integral transforms is given in Table 7.1.

Transform	**Domain**	**Core Strength**	**Key Limitation**	**Best Applications**
Laplace	$[0, \infty)$	Encodes initial conditions; reduces ODEs to algebraic equations.	Restricted to half-line; poor for variable coefficients.	IVPs, Control Systems, Linear ODEs
Fourier	$(-\infty, \infty)$	Exact frequency decomposition and spectral analysis.	Requires absolute integrability; lacks initial condition encoding.	BVPs, Wave Equations, Signal Processing
Mellin	$(0, \infty)$	Exploits scaling and multiplicative structures.	Narrow convergence strip cannot handle exponential growth.	Euler-type ODEs, Asymptotics, Fractals
Weierstrass	$(-\infty, \infty)$	Gaussian smoothing solves the heat equation directly.	Information loss suppresses high-frequency detail.	If there is diffusion and noise in the system

Table 7.1: Comparison of the different integral transforms.

7.1 Analysis of Strengths and Operational Limitations

There are three main limitations to consider for these transforms:

1. **Linearity**: All four transforms are linear operators. As a result, they cannot be applied directly to non-linear differential or integral equations (although they remain useful in perturbation methods or hybrid numerical schemes).

2. **Inversion Complexity**: Recovering the solution in the original domain often requires contour integration (Bromwich integral for Laplace) or numerical techniques. The process can be sensitive to pole locations and rounding errors.

3. **Growth and Convergence Restrictions**: Each transform is only applicable to functions satisfying specific growth or integrability conditions (exponential order for Laplace, absolute integrability for Fourier, power-law decay/growth for Mellin, sub-exponential growth for Weierstrass). Functions that violate these conditions (e.g. e^{t^2}) lie outside the scope of the transform.

7.2 Transform Selection Based on Problem Structure

The most effective transform is determined by the **structure** of the problem.

- If the system is governed by **time-shifts** or has a initial point, the **Laplace transform** is often the most appropriate tool.
- If the system is governed by **periodicity**, oscillation, or frequency content, the **Fourier transform** is the best choice.
- If the system uses **scaling**, self-similarity, or multiplicative structure, the **Mellin transform** is particularly effective.
- If the system is governed by **diffusion**, smoothing, or noise reduction, the **Weierstrass transform** is the most suitable.

Integral transforms should be selected according to the dominant behaviour of the problem: whether it is best understood through **shifting**, **frequency**, **scaling**, or **smoothing**. In this sense, integral transforms reveal the structure of functions, equations, and physical systems.

A Tables of Common Transforms

The Fourier transform does not have an RoC as it is defined on the frequency axis ($\omega \in \mathbb{R}$), with no adjustable real part. The kernel $e^{-\mathrm{i}\omega t}$ provides only oscillation, offering no damping. Convergence requires the integral to exist for **all** real frequencies — either it holds everywhere on the real line (for suitably integrable or square-integrable functions) or nowhere.

Table A.1: Common Laplace Transforms

Function $f(t)$ **(**$t \geq 0$**)**	$\mathcal{L}\{f(t)\}(s)$	**RoC**
1	$\frac{1}{s}$	$\Re(s) > 0$
t	$\frac{1}{s^2}$	$\Re(s) > 0$
t^n $(n \in \mathbb{N})$	$\frac{n!}{s^{n+1}}$	$\Re(s) > 0$
e^{at}	$\frac{1}{s-a}$	$\Re(s) > \Re(a)$
$\sin(at)$	$\frac{a}{s^2+a^2}$	$\Re(s) > 0$
$\cos(at)$	$\frac{s}{s^2+a^2}$	$\Re(s) > 0$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\Re(s) > -a$
$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	$\Re(s) > \Re(a)$
$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$	$\Re(s) > \Re(a)$
$\delta(t)$	1	all s
$u(t)$	$\frac{1}{s}$	$\Re(s) > 0$

Table A.2: Common Fourier Transforms

Function $f(t)$	$\mathcal{F}\{f(t)\}(\omega)$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$e^{-a\|t\|}$ $(a > 0)$	$\frac{2a}{a^2+\omega^2}$
$\text{rect}(t)$ $(\|t\| < 1/2)$	$\text{sinc}(\omega/2\pi)$
$e^{-\pi t^2}$	$e^{-\omega^2/4\pi}$
$e^{i\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$\frac{\pi}{i}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$u(t)$	$\pi\delta(\omega) + \frac{i}{\omega}$

Table A.3: Common Mellin Transforms

Function $f(t)$ **(**$t \geq 0$**)**	$\mathcal{M}\{f(t)\}(s)$	**RoC**
$t^{\alpha-1}$	$\Gamma(s+\alpha)$	$\Re(s+\alpha) > 0$
$t^{\alpha-1}e^{-at}$ $(a > 0)$	$a^{-s}\Gamma(s+\alpha)$	$\Re(s+\alpha) > 0$
$t^{\alpha-1}(\ln t)^k$	$\frac{d^k}{ds^k}\Gamma(s+\alpha)$	$\Re(s+\alpha) > 0$
e^{-t}	$\Gamma(s)$	$\Re(s) > 0$
e^{-t^β} $(\beta > 0)$	$\beta^{-s}\Gamma(s/\beta)$	$\Re(s) > 0$
$t^{\alpha-1}(1+t)^{-(\alpha+\beta)}$	$B(s,\beta)$	$\Re(s) > 0$, $\Re(\beta) > 0$
$t^{\alpha-1}(1+t)^{-\beta}$	$B(s,\beta-\alpha)$	$\Re(s) > 0$, $\Re(\beta-\alpha) > 0$
$\sin(at)$	$a^{-s}\Gamma(s)\sin(\pi s/2)$	$0 < \Re(s) < 1$
$\cos(at)$	$a^{-s}\Gamma(s)\cos(\pi s/2)$	$0 < \Re(s) < 1$
$t^{\alpha-1}\sin(at)$	$a^{-s}\Gamma(s+\alpha)\sin(\pi(s+\alpha)/2)$	$-\alpha < \Re(s) < 1-\alpha$
$t^{\alpha-1}\cos(at)$	$a^{-s}\Gamma(s+\alpha)\cos(\pi(s+\alpha)/2)$	$-\alpha < \Re(s) < 1-\alpha$
$(1+t^2)^{-\alpha/2}$	$2^{-\alpha}\pi^{1/2}\frac{\Gamma(\alpha/2)}{\Gamma(\alpha/2+1/2)} \cdot \frac{\Gamma(s+\alpha/2-1/2)}{\Gamma(s+\alpha/2)}$	$\Re(s) > 1-\alpha$
$J_\nu(at)$ (Bessel function)	$2^{s-1}a^{-s}\Gamma(\frac{s+\nu}{2})/\Gamma(\frac{-s+\nu+2}{2})$	$-\Re(\nu) < \Re(s) < 1/2$